Kartikey Pathak

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Education

Veermata Jijabai Technological Institute (VJTI) — Mumbai, India
B.Tech in Electrical Engineering, Minor in Robotics (CGPA: 8.04) — *2023 – 2027*

Experience

Eyecandy Robotics — April 2026 – August 2026
Electronics and Hardware Engineer — *Bangalore, India*

- Designed production PCBs for a legged consumer robot: Raspberry Pi Zero 2W and Milk-V compute carriers, servo driver and sensor boards, and a power board running 17 motors off one pack
- Owned schematic capture, routing and fabrication output, integrating IMU, servo bus, audio amplifier and microphone array with test points and solder-jumper options for hardware bring-up

NishCorp Technologies — July 2025 – November 2025
Robotics Control Systems Engineer Intern — *Hybrid*

- Developed Gazebo simulation for a robotic wheelchair platform to analyze motion and steering, applying differential drive kinematics and integrating IMU plugins
- Contributed to the half-scale hardware prototype through integration, testing and field evaluation

Krishna Defence and Allied Industries Limited — April 2025 – July 2025
Research Intern - Defence Technology — *IIT Gandhinagar*

- Developed firmware for Qorvo DWM3001CDK Ultra-Wideband modules using Zephyr RTOS for indoor positioning where GPS is unreliable, handling low-level drivers and RF packets
- Implemented SLAM and NAV2 for autonomous forklift POC navigation in warehouse environments
- Achieved precision localization error of 30 cm across a 10,000 sq meter area and 2 cm in a 50 sq meter area

Projects

PCB Axial-Flux BLDC Motor | *KiCAD, FEMM, Python* — Aug 2026 – Present

- Designing a coreless axial-flux motor whose stator is the PCB itself: 100 mm two-layer board, 6 programmatically generated wedge coils over 8 poles, dual rotor with 16 N42 magnets across a 0.5 mm air gap
- Built FEMM magnetostatic models to extract torque constant, back-EMF and torque ripple, identifying a 3.3x torque loss from the plastic rotor before fabrication

FOC Controller for Micro BLDC Motor | *STM32, DRV8311H, PCB Design* — Jan 2026 – Feb 2026

- Developed a Field-Oriented Control (FOC) system for a 4300 KV micro BLDC motor operating from a 3.7 V supply
- Designed and routed a custom PCB integrating an STM32 microcontroller, DRV8311H motor driver, and AS5600 encoder

Smart Switch Board | *ESP32, ESP-IDF, MQTT, BLE, Flutter, PCB Design* — 2025 – Present

- Built a network-controlled mains wall switch across every layer: CAD enclosure, a relay and optocoupler board isolating 220 V from a 3.3 V ESP32, ESP-IDF firmware, a TLS MQTT broker and a Flutter app
- Designed the physical switches to keep working with network, broker and app unavailable, with WiFi provisioning over BLE and switch state persisted in NVS across power cuts

MARIO — 3DOF Robotic Manipulator | *ROS2, Gazebo Fortress, ESP32, micro-ROS* — Apr 2025 – Aug 2025

- Developed ESP-IDF firmware and a micro-ROS pipeline for real-time hardware-software integration, migrating the simulation to Gazebo Fortress for teleoperation testing
- Manufactured, tested and distributed 50+ robotics kits used by 200+ students for manipulator control

Open Manipulator X Simulation | *ROS2, Gazebo, RViz, Computer Vision* — June 2024 – July 2024

- Programmed forward and inverse kinematics for an Open Manipulator X robot arm to perform precise pick-and-place tasks
- Implemented object detection and tracking algorithms to locate target objects in the Gazebo simulation environment

Automated Coin Sorter | *Raspberry Pi, TensorFlow, OpenCV, Servos* — 24-Hour Hackathon

- Classified Rs 1, 2, 5 and 10 coins from a camera feed using a CNN trained by transfer learning, reaching above 95% accuracy on Rs 5 and Rs 10 coins
- Routed each classified coin into its bin using servo-actuated gates driven from the Raspberry Pi

Competitions

eYantra Warehouse Drone Competition | *IIT Bombay* — Oct 2024 – Feb 2025

- Developed autonomous navigation and PID control algorithms for stable hovering and motion of a simulated drone
- Implemented ArUco marker detection using computer vision for autonomous localization and task execution

Positions of Responsibility

Society of Robotics and Automation (SRA) Aug 2024 – Mar 2026
Mechanical Head & Active Member *VJTI, Mumbai*

- Conducted technical workshops on Embedded C, ESP32 microcontrollers, and ROS-based systems for 200+ students
- Mentored junior teams in developing self-balancing and line-following robots, providing hardware/software expertise

Achievements

- Led the Electrical Engineering branch to its first overall win at Pratibimb, VJTI's cultural festival, in the competition's 26-year history
- Directed *Plot Se Panga*, a short film that won first prize across all branches at VJTI

Technical Skills

Programming Languages: C, C++, Python, Embedded C
Software & Frameworks: ROS2, Gazebo, RViz, ESP-IDF, STM32CubeIDE, KiCAD, micro-ROS
Hardware: ESP32, STM32, Raspberry Pi, UWB Modules, Motor Drivers, BLDC/Stepper/DC Motors, LiDAR
Core Competencies: Robot Kinematics, Field-Oriented Control, SLAM, NAV2, PID Control, PCB Design, IoT